Maximum likelihood estimation.

Def: Let $f(\underline{x};\theta)$ denote the joint pdf/pmf of $\underline{X}=(X_1,\dots,X_n)$. Then given $\underline{X}=\underline{x}$ is observed, the function $\mathcal{L}:\Theta\to[0,\infty)$ defined by

$$\mathcal{L}(\theta)=\mathcal{L}(\theta;\underline{x})=f(\underline{x};\theta)\qquad \theta\in\Theta$$

is called the likelihood function (CB Def 6.3.1)

We often consider $X_1,\dots,X_n$ to be a random sample, that is the X_i are independent and identically distributed.

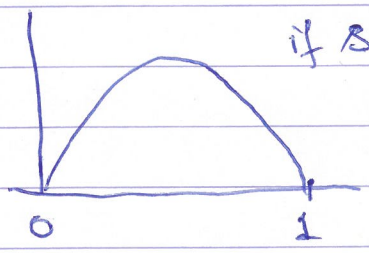
Example (Bernoulli) If $X_1,\dots,X_n\sim$ Bernoulli(p), the pmf of a single X_i is

$$p^x(1-p)^{1-x}\qquad p\in[0,1],\quad x\in\{0,1\}$$

we interpret $0^0=1$. The likelihood function $\mathcal{L}(p)$

$$\mathcal{L}(p;\underline{x})=\prod_{j=1}^{n}\left\{p^{x_j}(1-p)^{1-x_j}\right\}\qquad p\in[0,1]$$

$$=p^{s}(1-p)^{n-s}\qquad \text{where } s=\sum_{j=1}^{n}x_j.$$



Example (Exponential) If $X_1,\dots,X_n\sim\text{Exp}(\lambda)$, the pdf of a single X_i is

$$\lambda\exp(-\lambda x),\quad x>0,\quad \lambda>0.$$

The likelihood function is

$$\mathcal{L}(\lambda;\underline{x})=\prod_{j=1}^{n}\left\{\lambda\exp(-\lambda x_j)\right\}\qquad \lambda>0$$

$$=\lambda^n\exp(-\lambda s)\qquad s=\sum_{j=1}^{n}x_j.$$

If $\underline{X}$ is discrete, then $\mathcal{L}(\theta;\underline{x})=P_\theta(X=x)$. If $\theta_1,\theta_2\in\Theta$ such that $\mathcal{L}(\theta_1;x)>\mathcal{L}(\theta_2;x)$, then θ_1 would be plausible than θ_2.

Def: For each sample point x, let $\hat{\theta}(x)$ be a parameter value at which $\mathcal{L}(\theta; x)$ attains its maximum (as a function of θ with x fixed), i.e.

$$\mathcal{L}(\hat{\theta}(x); x) = \max_{\theta \in \Theta} \mathcal{L}(\theta; x)$$

A maximum likelihood estimator (MLE) of θ based on X is $\hat{\theta}(X)$. (CB Def 7.2.4)

Suppose $g: D \to \mathbb{R}$

- If D is closed and bounded, and g is continuous then the global maximum exists (Extreme value theorem)
- If g'' is continuous near c and $g'(c) = 0$ with $g''(c) < 0$, then g has local maximum at c.
- If $g'(x) > 0$ for all $x < c$ and $g'(x) \leq 0$ for all $x > c$, then c is global maximum.

* multidimensional case $\nabla g = 0$, then checking Hessian to see if we have local max.

It is often easier to work with the log likelihood function $\ell: \Theta \to \mathbb{R}$ defined by

$$\ell(\theta) = \log \mathcal{L}(\theta),$$

where log is the natural logarithm.

Example (Exponential) The likelihood function

$$\mathcal{L}(\lambda; x) = \lambda^n \exp(-\lambda s) \quad \text{where } s = \sum_{j=1}^{n} x_j, \ \lambda > 0.$$

The log likelihood function is

$$\ell(\lambda) = n \log \lambda - \lambda s, \quad \lambda > 0$$

To determine critical point (s), we differentiate

$$\ell'(\lambda) = \frac{n}{\lambda} - s$$

Upon setting $\ell'(\lambda) = 0$, we have one critical point $\hat{\lambda} = \frac{n}{s}$.

The first derivative test shows this is a global maximum. Or checking the second derivative

$$\ell''(\lambda) = \frac{-n}{\lambda^2} < 0 \quad \text{for all } \lambda > 0.$$

Hence, the maximum likelihood estimator is $\hat{\lambda}(X) = \frac{n}{S} = \frac{1}{\bar{X}}$ where $S = \sum_{j=1}^{n} X_j$.

Example (Poisson) Suppose $X_1, \ldots, X_n \sim \text{Poisson}(\lambda)$. The pmf of Poisson($\lambda$) is

$$\frac{e^{-\lambda}\lambda^x}{x!}, \quad x = 0, 1, 2, \ldots \qquad \lambda > 0.$$

The likelihood function is

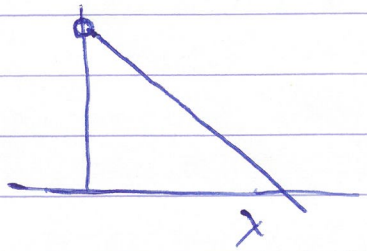
$$\mathcal{L}(\lambda; \underline{x}) = \prod_{j=1}^{n} \left\{ \frac{e^{-\lambda}\lambda^{x_j}}{x_j!} \right\} = \frac{\exp(-n\lambda)\,\lambda^{S}}{\prod_{j=1}^{n} x_j!}$$

where $S = \sum_{j=1}^{n} x_j$.

The log likelihood function is

$$\ell(\lambda) = -n\lambda + S\log(\lambda) - \sum_{j=1}^{n} \log(x_j!)$$

What does the log likelihood look like when $S = 0$.



As $\lambda \downarrow 0$ the log likelihood increases, but 0 is not in the parameter space.

Hence, the MLE does not exist when $S = 0$.

Practice: Find the MLE when $S > 0$.

Even when the MLE exists, it may not be unique.

Example (Laplace location family) Consider the pdf

$$\frac{1}{2}\exp(-|x-\theta|), \quad x \in \mathbb{R}, \quad \theta \in \mathbb{R}.$$

If $X_1, \ldots, X_n$ is a random sample from this distribution, then the likelihood function is

$$\mathcal{L}(\theta; \underline{x}) = \prod_{j=1}^{n} \left\{ \frac{1}{2}\exp(-|x_j-\theta|) \right\} = 2^{-n}\exp\left(-\sum_{j=1}^{n}|x_j-\theta|\right)$$

The log likelihood function is

$$\ell(\theta) = -n\log(2) - \sum_{j=1}^{n}|x_j-\theta| \qquad \theta \in \mathbb{R}.$$

Suppose we minimise the function

$$h(\theta) = \sum_{j=1}^{n}|x_j-\theta|$$

Assume $n = 2m$ (ie n even). Take θ' and $\theta'+\varepsilon$ in $(x_{(m)}, x_{(m+1)})$ – from m-th smallest to m+1-smallest obs.

~~$h(\theta') = 2$~~ $\quad h(\theta) = \sum_{j=1}^{n}|x_{(j)}-\theta|$

$$h(\theta') = \sum_{j=1}^{m}(\theta' - x_{(j)}) + \sum_{j=m+1}^{n}(x_{(j)} - \theta')$$

$$h(\theta'+\varepsilon) = \sum_{j=1}^{m}(\theta'+\varepsilon - x_{(j)}) + \sum_{j=m+1}^{n}(x_{(j)} - \theta' - \varepsilon)$$

$$= \sum_{j=1}^{m}(\theta' - x_{(j)}) + m\varepsilon + \sum_{j=m+1}^{n}(x_{(j)} - \theta') - m\varepsilon.$$

$$= h(\theta')$$

Example (Cauchy location family) Consider the pdf

$$\frac{1}{\pi(1+(x-\theta)^2)} \qquad x \in \mathbb{R},\ \theta \in \mathbb{R}.$$

If we have a random sample from this distribution, the likelihood function is

$$\mathcal{L}(\theta; \underline{x}) = \prod_{j=1}^{n} \frac{1}{\pi(1+(x_j-\theta)^2)} \qquad \theta \in \mathbb{R}.$$

and the log likelihood function is

$$\ell(\theta) = -\sum_{j=1}^{n} \log(1+(x_j-\theta)^2) - n\log\pi.$$

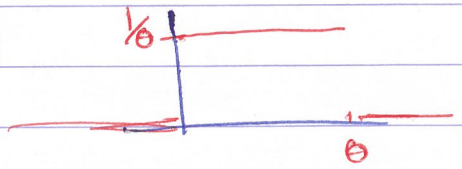
This log likelihood can have several local max & min.

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Example (Uniform) The uniform distribution on $[0,\theta]$ has pdf is

$$\theta^{-1}\, \mathbb{I}(0 \le x \le \theta)$$

$\theta > 0$



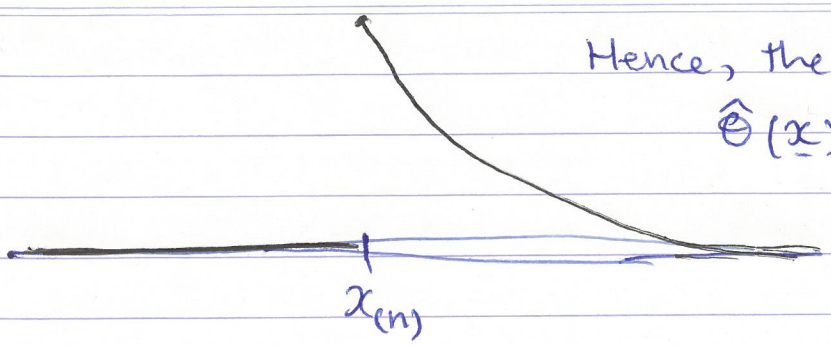
If we have a random sample ~~from~~ $X_1, \ldots, X_n$ from $U[0,\theta]$, the likelihood is

$$\mathcal{L}(\theta; x) = \prod_{j=1}^{n} \left\{\theta^{-1}\, \mathbb{I}(0 \le x_j \le \theta)\right\}$$

$$= \theta^{-n} \prod_{j=1}^{n} \mathbb{I}(0 \le x_j \le \theta)$$

$$= \theta^{-n}\, \mathbb{I}(0 \le x_{(1)})\, \mathbb{I}(x_{(n)} \le \theta)$$

where $x_{(1)}$ is the smallest obs and $x_{(n)}$ is the largest obs.



Hence, the MLE is $x_{(n)}$

$$\hat{\theta}(\underline{x}) = x_{(n)}$$

Example (Normal distribution) Suppose $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$. $\theta = (\mu, \sigma^2)$ and $\Theta = \mathbb{R} \times (0, \infty)$. The likelihood function is

$$\mathcal{L}(\theta; \underline{x}) = \prod_{j=1}^{n} \left\{ (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{1}{2\sigma^2} (x_j - \mu)^2 \right) \right\}$$

$$= (2\pi\sigma^2)^{-n/2} \exp\left(\frac{-1}{2\sigma^2} \sum_{j=1}^{n} (x_j - \mu)^2 \right)$$

The log likelihood is

$$\ell(\theta) = \frac{-n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{j=1}^{n} (x_j - \mu)^2 - \frac{n}{2} \log(2\pi)$$

If we fix σ^2, the $\ell(\theta)$ is a quadratic function in μ

Differentiating wrt μ

$$\frac{\partial \ell}{\partial \mu} = \frac{+1}{\sigma^2} \sum_{j=1}^{n} (x_j - \mu)$$

So $\frac{\partial \ell}{\partial \mu} = 0$ has only one solution,

$$\frac{1}{\sigma^2} \sum_{j=1}^{n} (x_j - \mu) = 0 \Rightarrow \sum_{j=1}^{n} x_j = n\mu \Rightarrow \hat{\mu} = \frac{1}{n} \sum_{j=1}^{n} x_j$$

This is global maximum since

$$\frac{\partial^2 \ell}{\partial \mu^2} = \frac{-n}{\sigma^2} < 0 \quad \text{for all } \mu \in \mathbb{R}.$$

So we can say for all σ^2

$$\mathcal{L}(\mu, \sigma^2) \leq \mathcal{L}(\bar{x}, \sigma^2)$$

Related to profile likelihood (CB 7-5-5)

We now maximise $\ell(\bar{x}, \sigma^2)$ over σ^2.

$$\ell\ell(\bar{x}, \sigma^2) = -\frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{j=1}^{n}(x_j - \bar{x})^2$$

Differentiate wrt σ^2

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{j=1}^{n}(x_j - \bar{x})^2$$

So the critical point(s) satisfy

$$\frac{-n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{j=1}^{n}(x_j - \bar{x})^2 = 0$$

$$\Rightarrow \quad \frac{1}{\sigma^2}\sum_{j=1}^{n}(x_j - \bar{x})^2 = n \quad \Rightarrow \quad \sigma^2 = \frac{1}{n}\sum_{j=1}^{n}(x_j - \bar{x})^2$$

Again the first derivative test shows this is a global maximum

The MLE for $N(\mu, \sigma^2)$ is $\left(\bar{X}, \frac{1}{n}\sum_{j=1}^{n}(X_j - \bar{X})^2\right)$.

Question: Does it matter that we parametrised the distribution in terms of σ^2 rather than σ?

Invariance property of MLE

We have a parametric model $\mathcal{F} = \{f(\cdot;\theta) : \theta \in \Theta\}$. We want to estimate $\Psi(\theta)$. The invariance property of MLE says that if $\hat{\theta}$ is the MLE of θ, then for any function $\Psi : \Theta \to \Psi$, the MLE of $\Psi(\theta)$ is $\Psi(\hat{\theta})$. (CB Thm 7-2-10)

Suppose Ψ is a one-to-one transformation from Θ to Ψ. $\mathcal{L}(\theta)$ - the likelihood on Θ and $\mathcal{L}^*(\Psi)$ - the likelihood on Ψ. Then for any $\Psi_0 \in \Psi$

$$\mathcal{L}^*(\Psi_0) = \mathcal{L}(\Psi^{-1}(\Psi_0))$$

Hence, $\mathcal{L}^*(\Psi(\hat{\theta})) = \mathcal{L}(\hat{\theta})$ which is the maximal value.

When ψ is not one-to-one, this property is extended by introducing the induced likelihood

$$\mathcal{L}^*(\psi_0) = \sup_{\{\theta : \psi(\theta) = \psi_0\}} \mathcal{L}(\theta).$$

So the maximiser of the induced likelihood $\mathcal{L}^*$ is $\hat{\psi} = \psi(\hat{\theta})$, where $\hat{\theta}$ is the MLE of θ.

Example (Normal) The MLE of σ is $\sqrt{\frac{1}{n}\sum_{j=1}^{n}(X_j - \bar{X})^2}$.